

Programming for Data Science One-Day Open-Ended Hackathon

From Problem to Data Science Workflow

Date	Sunday, 10 May 2026	Team size	1–2 students
Submission	one PDF report only	Deadline	5:00 PM IST

Final submission deadline: 5:00 PM IST on Sunday, 10 May 2026.

Submit **exactly one elaborative PDF report** before this deadline. There is no live presentation requirement and no separate file package. Any website, dashboard, repository, notebook, dataset, or prototype link must be embedded clearly inside the PDF report. The submission timestamp on the course portal is the official timestamp.

1 Purpose

This hackathon is not a fixed-dataset exercise. Your team must begin with an open-ended real-world problem, convert it into a clear programming and data science challenge, and design a solution workflow using concepts learned so far in the course.

Many students in this course are working professionals. You are encouraged to choose a problem that connects to your workplace, professional domain, field experience, or an operational setting you understand well. The report should make the business or institutional context explicit while avoiding confidential, personally identifiable, or proprietary information.

The main goal is to practice the most important skill in applied data science:

turning an unstructured problem into a technically valid, reproducible, decision-oriented workflow.

You are expected to think like a data scientist and an engineer. A good submission should make it clear what decision is being supported, what data is needed, what method is appropriate, how the result will be validated, and how the workflow could be reproduced or deployed.

2 Core Challenge

Hackathon Task

In a team of one or two, identify a real-world problem, articulate the business case, formulate the programming challenge, and design a complete flowchart for a data science solution. Your final output is one detailed PDF report that contains the business case, technical workflow, method justification, proof-of-concept evidence, and any embedded website or prototype links.

Your project can come from any domain, including but not limited to:

- transport, logistics, and delivery systems;
- campus operations, student services, or facility management;
- water, flooding, climate, air quality, or urban infrastructure;
- health services, public services, or emergency response;
- agriculture, energy, finance, education, or customer support.

Data Expectation

Use of real or realistic data is strongly encouraged. Open-source datasets, public APIs, government portals, published benchmark datasets, or properly anonymized workplace-derived summaries are suitable choices. If real data are unavailable, confidential, too sparse, or impractical to access within one day, you may generate synthetic data. Synthetic data must be clearly labelled as synthetic, and the report must explain what assumptions were used to generate it and which parts of the business case it approximates.

3 Team Rules

- Teams may have **1–2 students**.
- All team members must contribute to the business case, problem framing, technical workflow, proof-of-concept evidence, and written report.
- Teams larger than two require instructor approval before **10:00 AM IST**.
- Each team submits exactly one final PDF report.
- You may use notes, documentation, internet resources, and AI coding assistants. You remain responsible for correctness, originality, and explanation.

4 Only Required Submission

Each team must submit **one self-contained, elaborative PDF report**. This report is the only required submission and the main grading artifact. A suggested length is **8–12 pages**, excluding optional appendices, but clarity and completeness matter more than page count.

The report must contain all important material directly inside the PDF:

1. **Business case** – the stakeholder, pain point, decision, value, and success metric.
2. **Problem statement** – the real-world problem converted into a data/programming challenge.
3. **Solution flowchart** – the central technical artifact of the hackathon.
4. **Method justification** – explain why the selected statistical, ML, data-ingestion, dashboard, database, or reproducibility methods fit the problem.
5. **Proof-of-concept evidence** – screenshots, code snippets, outputs, tables, plots, or a concise explanation of the working step.
6. **Embedded links** – if a website, dashboard, repository, notebook, dataset, API, or prototype is developed or used, include a clickable link in the report with access notes.

Single-PDF Rule

Do not submit a zip file, slide deck, separate notebook, separate flowchart image, or separate README as the primary submission. If you create those artifacts, link to them from inside the PDF report and explain exactly what the evaluator should see.

5 Required Report Structure

Use the following structure for the final PDF report. You may add subsections if needed, but do not omit the required sections.

1. **Cover page**: team member names, roll numbers, project title, course title, date, and one-sentence project summary.

2. **Executive summary:** 150–250 words explaining the problem, proposed solution, expected value, and most important limitation.
3. **Business case:** define the user, current pain point, decision to be improved, why the problem matters, expected benefit, cost/risk of wrong decisions, and success metric.
4. **Problem formulation:** convert the business problem into a data science task such as prediction, classification, hypothesis testing, optimization, scraping, streaming, database serving, or dashboarding.
5. **Data and programming challenge:** describe data sources, expected schema, ingestion method, cleaning needs, technical obstacles, and assumptions. State whether the data are real, open-source, anonymized workplace-derived, realistic mock data, or synthetic.
6. **Solution flowchart:** include a readable flowchart showing the full path from problem to data to method to decision output.
7. **Methodological justification:** explain the statistical, ML, scraping/API, Kafka/database, Docker, or optimization choices and why they are appropriate.
8. **Technical method:** describe the specific implementation logic, model, test, metric, objective function, database/dashboard design, or reproducibility plan.
9. **Proof-of-concept evidence:** show at least one working technical step through code snippets, screenshots, plots, command output, model output, or dashboard screenshot.
10. **Website or prototype link, if applicable:** embed the clickable URL and state whether it is a live site, local demo, GitHub Pages app, Streamlit app, notebook, repository, or mockup. Include screenshots so the report remains understandable even if the link fails.
11. **Validation, risks, and limitations:** explain how the solution would be checked, where leakage, confounding, or bias could enter, and what should not be claimed.
12. **Implementation plan:** describe the next 2–3 technical steps required to turn the design into a more complete working system.
13. **References and AI-use note:** cite datasets, APIs, websites, documentation, code snippets, and AI prompts/tools used.

6 Business Case Requirement

The business case must be clearly carved out as a dedicated section. It should not be a vague motivation paragraph. It must answer:

- Who owns the problem and who will use the output?
- What operational, financial, safety, environmental, service-quality, or decision-making pain is being addressed?
- What decision will change because of the workflow?
- What is the baseline or current way of making the decision?
- What measurable improvement would count as success?
- What are the risks of false positives, false negatives, biased data, or wrong interpretation?
- What is the smallest deployable version of the solution?

7 Data Source Requirement

Your report must state the data basis of the project. Use the strongest feasible option:

1. **Real or open-source data:** preferred. Include the source, access link, license or citation if available, and the fields used.
2. **Anonymized workplace-derived data:** acceptable if it does not expose confidential, proprietary, or personally identifiable information. Aggregate or mask sensitive fields.
3. **Realistic mock data:** acceptable when the structure is realistic but values are invented for privacy or access reasons.
4. **Synthetic data:** acceptable as a last resort. Explain how it was generated, what assumptions it encodes, and what conclusions cannot be made from it.

The use of synthetic data is not a penalty if it is transparent, technically justified, and connected to a realistic business case. It is a problem only if synthetic data are presented as real data or used to make unsupported claims.

8 Proof-of-Concept Requirement

The proof-of-concept does not need to be a complete product. It must, however, run or demonstrate one meaningful technical step. Acceptable examples include:

- loading and cleaning a sample dataset;
- scraping repeated units from a web page or calling an API;
- running a SciPy test, bootstrap interval, interpolation, or optimization;
- building a leakage-safe scikit-learn pipeline;
- choosing and computing a suitable evaluation metric;
- simulating a Kafka-like event stream or storing latest state in a database;
- creating a simple dashboard, CLI output, or decision report;
- writing a Dockerfile or Compose file for reproducibility.

Evidence of this proof-of-concept must appear inside the PDF report. A link alone is not enough; include at least one screenshot, code snippet, plot, output table, or explanation that makes the technical work visible in the report.

9 Website and Link Embedding

If your team develops a website, dashboard, hosted notebook, GitHub repository, or web app, the link must be embedded directly inside the PDF report. Use a clear label such as:

Prototype website: `https://...`

For every embedded link, include:

- what the link opens: website, dashboard, repository, notebook, dataset, API, or documentation;
- whether login is required;
- what the evaluator should click or inspect;
- one screenshot or output snapshot inside the PDF;
- fallback notes if the link is local-only, temporary, or likely to expire.

The PDF must still be understandable even if an external link is unavailable during evaluation.

10 Methodological Justification

Do not add methods merely because they were covered in the course. Choose only the methods that the problem actually requires, and justify them in professional terms. For example, a report may justify a confidence interval because the decision-maker needs uncertainty, a recall-focused classifier because false negatives are costly, an API because it is more stable than scraping HTML, a database because a dashboard needs queryable latest state, or Docker because the workflow must be reproducible.

The evaluator should be able to see why each technical choice is necessary for the business problem, not just that a course concept was mentioned.

11 Required Problem Framing

Your problem statement must answer the following questions:

1. **What is the real-world problem?**
2. **Who is the decision-maker or user?**
3. **What decision will your workflow support?**
4. **What data would be needed?**
5. **What is the programming challenge?**
6. **What could go wrong if the analysis is done poorly?**
7. **How will you validate whether your solution is useful?**

11.1 Examples of Good Problem Formulation

- Weak: “We will analyze traffic data.”
- Strong: “We will help a campus transport manager decide when to add a shuttle by predicting high-wait periods, estimating uncertainty in average wait reduction, and optimizing a deployment threshold.”
- Weak: “We will use machine learning for healthcare.”
- Strong: “We will help a clinic prioritize appointment reminders by predicting no-show risk, using recall as the main metric because missing high-risk patients is costly.”

12 Flowchart Requirements

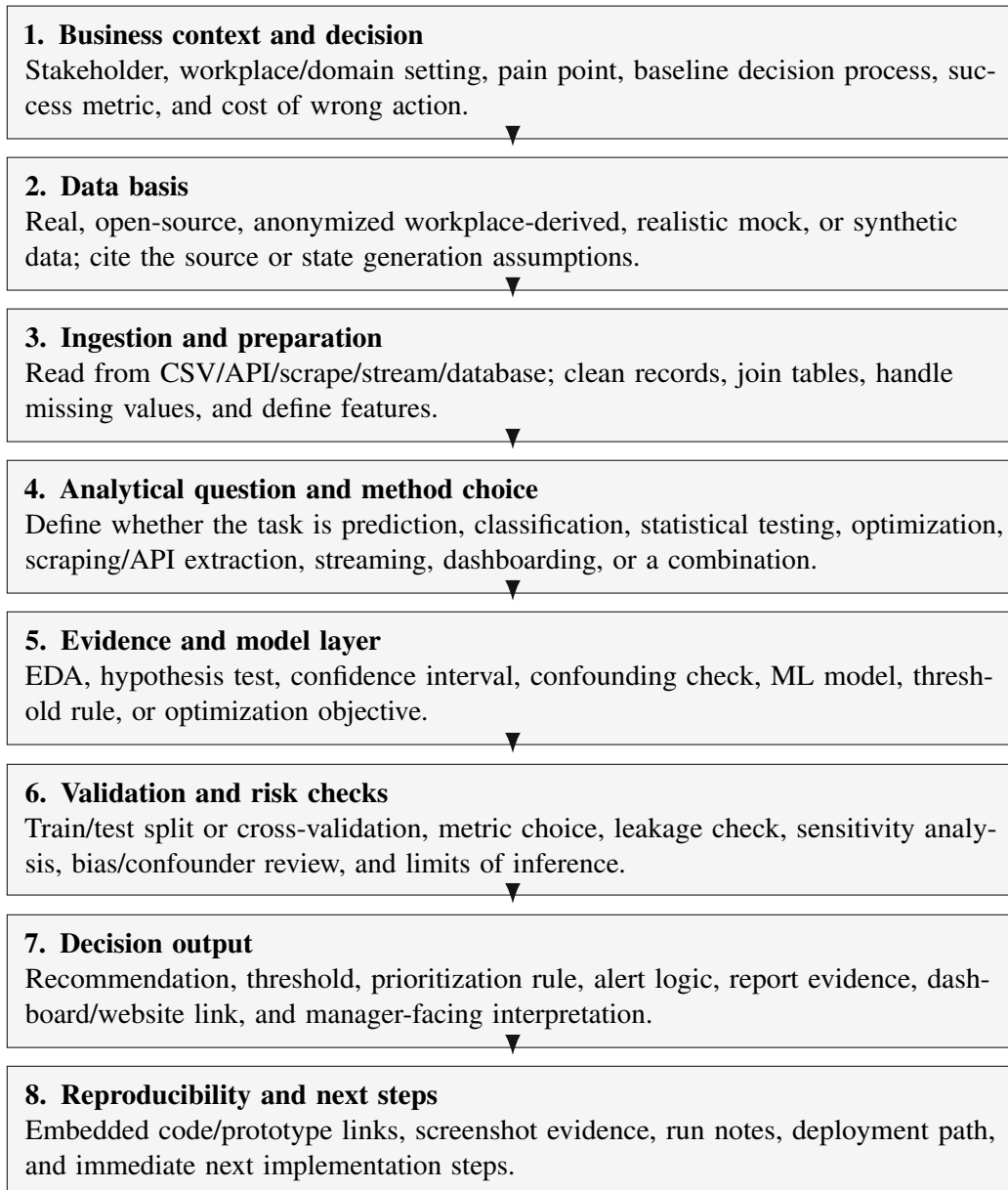
Your flowchart is the most important deliverable. It must be readable, technically specific, and connected to the course.

Flowchart Must Include

- Problem and decision objective.
- Data source or data generation plan, with a clear real, open-source, anonymized, mock, or synthetic label.
- Data ingestion method: CSV, API, scraping, stream, database, or simulation.
- Cleaning and preprocessing steps.
- Exploratory analysis and statistical reasoning.
- Model, optimization, or decision rule.
- Validation plan and evaluation metric.
- Final output: recommendation, website, dashboard, alert, report section, database table, or deployable script.
- Reproducibility plan: embedded link to code/notebook/site, environment notes, Docker plan, or database schema.

12.1 Example Flowchart

The example below is deliberately generic. Your team's version should replace each box with project-specific content, such as the stakeholder, data table names, target variable, metric, model, threshold, website link, and decision recommendation.



13 Hackathon Schedule

All times are in IST on Sunday, 10 May 2026.

Time	Activity and expected output
09:30–10:00	Briefing, team formation, rules, and judging criteria. Teams of 1–2 students must be finalized.
10:00–10:45	Problem discovery. Choose domain, workplace or public context, user, decision, and stakes. Draft the problem statement.
10:45–11:30	Convert the problem into a data and programming challenge. Identify real/open-source/anonymized/mock/synthetic data options, input/output, and technical obstacles.
11:30–12:30	Method selection. Decide which technical methods are appropriate and why.
12:30–13:30	Lunch / independent work.
13:30–14:45	Build the solution flowchart. Make every step technically specific.
14:45–15:45	Minimal proof-of-concept. Run one meaningful part of the workflow.
15:45–16:30	Prepare the final elaborative PDF report, including business case, flowchart, evidence, and embedded links.
16:30–17:00	Final checks: open the PDF, test embedded links, verify screenshots/flowchart readability, and confirm the filename.
17:00	Final submission deadline. Submit exactly one PDF report.

14 Submission Instructions

Submit exactly one PDF report by 5:00 PM IST on Sunday, 10 May 2026. Late edits are not counted for evaluation unless explicitly requested by the instructor.

Submit only one PDF file through the course submission portal. Use the filename:

PDS_Hackathon_TeamName_Roll1_Roll2.pdf

Do not upload a zip file, folder, separate code file, separate image, separate slide deck, or repository link as the primary submission. Put all links and evidence inside the PDF. If the portal is unavailable, email the PDF to the instructor before the deadline and keep evidence of the timestamp.

15 Evaluation Rubric

The hackathon will be evaluated out of 100 marks.

Criterion	Marks	What is expected
Business case and problem articulation	25	Clear stakeholder, pain point, baseline decision process, value proposition, decision to be improved, success metric, and risk of wrong decisions.
Programming and data challenge	15	Specific input data, data type, source or generation plan, ingestion challenge, cleaning needs, assumptions, and technical obstacles.
Methodological justification	15	Technical choices are appropriate for the problem and justified clearly. Methods may include statistical reasoning, ML, scraping/API, Kafka/database, Docker, optimization, or dashboarding, but only where useful.
Solution flowchart	15	Complete, logical, implementable workflow from business problem to data to method to decision output.
Proof-of-concept and embedded links	15	At least one meaningful technical step is evidenced inside the PDF. Website, dashboard, notebook, or repository links are clickable and clearly explained if used.
Validation, risks, and limitations	10	Clear validation plan, metric choice, leakage/confounding/bias discussion, and honest statement of what cannot be claimed.
Report quality and self-contained submission	5	The single PDF is well organized, readable, properly named, cited, and understandable without separate files.

16 Suggested Topic Seeds

You may choose your own topic. If you are stuck, use one of these as a starting point:

- Predicting campus shuttle crowding and deciding when to add an extra trip.
- Identifying high-risk customer support tickets and deciding when to assign priority support.
- Monitoring rainfall or drainage complaints and recommending inspection priority.
- Scraping public event listings and recommending resource allocation for campus services.
- Predicting late food deliveries and optimizing a dispatch threshold.
- Detecting unusual energy use in buildings and generating maintenance alerts.
- Designing a stream-processing workflow for sensor events and dashboard-ready state.
- Comparing two operational policies while accounting for confounding.
- Finding whether an overall trend reverses after grouping by location, severity, or user type.

17 Quality Checklist

Before submitting, check the following:

- Is the submission exactly one PDF file?
- Is the file named PDS_Hackathon_TeamName_Roll1_Roll2.pdf?
- Does the report contain a dedicated business case section?
- Does the project identify a real decision-maker?

- Is the decision specific enough to evaluate?
- Is the data source or data plan realistic?
- If synthetic data are used, are the generation assumptions and limits clearly stated?
- Is the programming challenge clearly stated?
- Does the flowchart go from input to output without missing major steps?
- Are the technical methods selected and justified correctly?
- Is at least one technical step demonstrated inside the PDF through code, screenshot, plot, table, or output?
- If a website, dashboard, notebook, or repository was developed, is the clickable link embedded and explained inside the PDF?
- Are limitations stated honestly?
- Can the evaluator understand the workflow and evidence without opening separate files?
- Was the final PDF report submitted before **5:00 PM IST**?

18 Academic and Professional Expectations

- Do not submit a generic AI-generated idea without technical detail.
- Cite external datasets, APIs, documentation, and code snippets where relevant.
- If AI tools are used, include a brief note listing the tool and the most important prompts.
- Do not claim a method is causal unless the design supports a causal claim.
- Do not report model accuracy alone if another metric is more appropriate for the decision.
- Do not ignore confounders, group-wise reversals, or leakage risks when they are relevant.

19 Final Reminder

Hard deadline: 5:00 PM IST, Sunday, 10 May 2026.

Submit exactly one final PDF report before the deadline. There is no live presentation component and no separate file package. A clear business case, technically honest workflow, embedded evidence, and reproducible links are more valuable than an ambitious but vague idea.